

ARNAV SINGH

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SUMMARY

Final-year Computer Engineering student (CGPA: 8.45) with hands-on experience building and deploying end-to-end machine learning systems, from data pipelines and model training through to production deployment. My work spans LLM fine-tuning, rigorous model evaluation, and fairness auditing, and I care more about proving a model actually works than simply getting it to run. This focus on evaluation over quick wins is what draws me toward LLM benchmarking as an area I want to grow in.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: Scikit-learn, XGBoost, SMOTE, ADASYN, Feature Engineering, SHAP, Conformal Prediction, MLflow, Weights & Biases

Deep Learning & NLP: CNN, Transformers, BERT fine-tuning, Mistral-7B, LoRA, QLoRA, 4-bit Quantization, Hugging Face, FAISS, BERTScore, Prompt Engineering

Data Analytics: Power BI, Excel, EDA, Cohort Analysis, A/B Testing, Hypothesis Testing, statsmodels, Matplotlib

MLOps & Deployment: FastAPI, Docker, Render, AWS (SageMaker basics), Git

PROJECTS

AI-Based Fake Identity & Document Screening System *Python, OpenCV, EasyOCR/Tesseract, DeepFace/InsightFace, Streamlit* [Project Link](#)

- Built a five-stage pipeline — OCR → Validation → Tamper Detection → Face Verification → Risk Scoring — using only pretrained models and classical CV, no training required.
- Implemented ICAO 9303 MRZ parsing with check-digit computation, rule-based field validation (dates, document-number regex, ISO nationality codes), and Error Level Analysis with heatmap visualization to localize tampered image regions.
- Combined DeepFace/InsightFace face-embedding similarity with a tunable weighted risk-scoring model to produce real-time PASS/FLAG verdicts; deployed as an interactive Streamlit dashboard.

Credit Card Fraud Detection *Python, XGBoost, SMOTE, SHAP, FastAPI, Render, MLflow* [Live Demo](#); [Project Link](#)

- Engineered fraud detection pipeline (SMOTE + XGBoost) achieving AUC-PR of 0.9752, 87.5% recall, and 0.152% false alarm rate, saving ₹5,338 versus a no-detection baseline.
- Used SHAP for global and transaction-level interpretability; documented feature importance findings with business-readable explanations.
- Deployed as a live REST API on Render with a real-time inference endpoint.

LegalSimplify - Legal Clause Simplifier (Mistral-7B) *LoRA, QLoRA, BitsAndBytes, Hugging Face* [Project Link](#)

- Fine-tuned Mistral-7B-Instruct using LoRA and 4-bit quantization on a custom legal dataset, reducing training memory by ~75%.
- Ran an ablation study across LoRA ranks (4, 8, 16, 32, 64) measuring perplexity, inference latency, and task quality; identified rank 16 as optimal for the legal domain.

Legal Query Classifier (BERT) *BERT, Docker, FastAPI* [Project link](#)

- Fine-tuned BERT for multi-task legal classification achieving 90%+ accuracy; addressed class imbalance using weighted loss.
- Evaluated using precision, recall, and F1-score per class; containerized with Docker and served via FastAPI for production-ready inference.

Customer Churn & Cohort Retention Analysis *SQL, Python, Power BI, statsmodels* [Project Link](#)

- Analyzed 540K+ transactions using cohort analysis, CTEs, and window functions to identify early-stage churn as the primary revenue risk.
- Ran hypothesis tests (chi-square, t-test) to validate churn signal significance across cohorts, informing retention strategy recommendations.

EDUCATION

B.Tech, Computer Engineering - ABES Engineering College, AKTU

2023 - 2027

CGPA: 8.45 | Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Probability & Statistics, Machine Learning

ACHIEVEMENTS & CERTIFICATIONS

- Solved 300+ DSA problems across [LeetCode](#) and [CodeChef](#); CodeChef Rating 1614 (3 Star).
- 2nd Place, ECell Eureka Competition Zonals (IIT Bombay ECell); 3rd Place, Retina AI Hackathon 2026.
- [2025 Google Solution Challenge Participant](#); [Google 5-Day Agentic AI Vibe Coding Contributor](#).
- Certifications: Master Regression & Prediction with Pandas and Python; Data Science and Machine Learning Fundamentals; SQL & SQL for Data Analytics; Data Fundamentals (IBM SkillsBuild).

LEADERSHIP & ACTIVITIES

Growth & Content Strategist - Aasira (STEM Beyond Labels)

Jul 2024 - Feb 2026

- Grew audience engagement by 35% through targeted social media campaigns; executed 10+ content pieces aligned with brand messaging, tracking performance via weekly dashboards.

Volunteer - GDGoC Member

Oct 2024 - Sep 2025

- Drove a 40% increase in event attendance and 30% improvement in content engagement through data-driven campaign optimization.